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The Impact of Prices on Analyst Cash Flow Expectations: Reconciling Subjective Beliefs Data with Rational Discount Rate Variation

Contents

1	Big picture	3
2	Data and measurement	3
2.1	Analyst cash-flow expectations	3
2.2	Annual EPS expectation revisions	3
2.3	Forecast errors and revisions	4
2.4	Staggered analyst timing	4
2.5	BMI instrument	4
2.6	FIT instrument	4
2.7	Stock-level controls and samples	5
3	Models and empirical specifications	5
3.1	BMI two-stage least squares	5
3.2	FIT two-stage least squares	5
3.3	Term structure of annual EPS expectations	5
3.4	Lagged FIT and reversal tests	6
3.5	Covariance decomposition	6
3.6	Rational-investor model	6
4	Identification and empirical design	6
4.1	What identifies the price effect	6
4.2	Threats addressed in the BMI design	6
4.3	Threats addressed in the FIT design	7
4.4	Why the result matters for subjective beliefs data	7
4.5	Limits of the design	7
5	Tables and figures	8
5.1	Figures 1–2: timing of analyst revisions and Russell reconstitution	8
5.2	Tables 1–2: Russell BMI instrument and baseline IV result	8
5.3	Figure 3: BMI robustness checks	9
5.4	Tables 3–4: FIT instrument and baseline IV result	9
5.5	Figures 4–5: term structure and persistence	10
5.6	Figure 6 and Table 5: FIT robustness and Rotemberg weights	10
5.7	Figure 7: covariance decomposition	11
5.8	Table 6: model estimation	11
6	Short summary	12
7	Conclusion	13
7.1	Core conclusion	13
7.2	Economic intuition	13

7.3 Takeaway	13
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1 Big picture

- **Question.** Can subjective beliefs data be reconciled with asset pricing models in which investors have rational expectations and discount-rate variation drives prices?
- **Main empirical claim.** Prices affect analyst cash-flow expectations. Exogenous price increases unrelated to cash-flow news raise analysts' earnings expectations and forecast errors.
- **Why this matters.** Prior work interprets the correlation between analyst cash-flow expectations, prices, forecast errors, and future returns as evidence that biased investor beliefs distort prices.
- **Alternative interpretation.** Analysts may learn from prices. If prices contain discount-rate variation as well as cash-flow information, analysts who learn from prices can form biased cash-flow expectations even when investors are rational.
- **Two instruments for price.** The paper uses two complementary instruments: changes in benchmarking intensity around Russell reconstitutions and flow-induced trading from mutual fund flows.
- **BMI result.** An exogenous 1% price increase from Russell benchmarking-intensity changes raises annual EPS expectation revisions by about 41 basis points.
- **FIT result.** An exogenous 1% price increase from mutual-fund flow-induced trading raises long-term growth expectations by about 5 basis points and annual EPS expectations by about 21 basis points.
- **Persistence.** The impact of prices on expectations does not quickly reverse; it persists in the term structure of EPS expectations and lagged FIT estimates.
- **Covariance decomposition.** The impact of prices on analyst expectations explains roughly a large share, about half in the author's discussion, of the covariance between prices and analyst expectations/forecast errors.
- **Model contribution.** A model with rational investors, private information, discount-rate variation, and analysts learning from prices can match key subjective-belief moments.
- **Takeaway.** Biased analyst expectations need not imply biased investor expectations. Subjective beliefs data may reflect analysts learning from prices that contain rational discount-rate variation.

2 Data and measurement

2.1 Analyst cash-flow expectations

- The paper uses analyst forecasts from I/B/E/S.
- The main short-horizon cash-flow expectations are annual EPS forecasts for one to four years ahead.
- The main long-horizon expectation is the long-term earnings growth (LTG) forecast.
- Expectations are measured at the analyst-institution, stock, and time level.

Interpretation: Analysts release reports on different dates, which allows the paper to measure expectation revisions between two reports for the same analyst and stock.

2.2 Annual EPS expectation revisions

For analyst institution a , stock n , quarter t , and horizon h , the annual EPS expectation revision is measured as

$$\Delta E_{a,n,t+h|t+1} = \frac{E_{a,n,t+h|t+1} - E_{a,n,t+h|t}}{E_{a,n,t+h|t}}. \quad (1)$$

The paper drops observations with nonpositive denominators and uses horizons $h = 1, 2, 3, 4$.

Interpretation: The normalization turns forecast changes into percentage revisions. Positive revisions mean analysts raise cash-flow expectations.

2.3 Forecast errors and revisions

For annual EPS forecasts,

$$ForecastError_{a,n,t+h|t} = RealizedEPS_{n,t+h} - E_{a,n,t+h|t}, \quad (2)$$

$$\Delta ForecastError_{a,n,t+h|t+1} = E_{a,n,t+h|t} - E_{a,n,t+h|t+1}. \quad (3)$$

Interpretation: The change in forecast error is the negative of the expectation revision. If exogenous price increases raise expectations without raising future earnings, forecast errors become more negative.

2.4 Staggered analyst timing

Analysts release reports on different dates. The price change used for a revision is the price change between the old and new report dates:

$$\Delta p_{a,n,t} = \frac{P_{n,\text{new report}} - P_{n,\text{old report}}}{P_{n,\text{old report}}}. \quad (4)$$

Interpretation: Staggered timing is important because it lets the author align price changes with exactly the period over which an analyst revises beliefs.

2.5 BMI instrument

The Russell benchmarking-intensity instrument uses mechanical changes in index benchmarking after Russell reconstitutions. The benchmarking intensity measure is

$$BMI_{n,t} = \sum_{j \in Index} \frac{InstitutionalAUM_{j,t} \times Weight_{n,j,t}}{MarketValue_{n,t}}. \quad (5)$$

The instrument is the June change in BMI for stocks near Russell market-cap cutoffs:

$$\Delta BMI_{n,t}. \quad (6)$$

Interpretation: Russell index assignment changes benchmarking demand and price pressure, but is conditionally unrelated to June cash-flow news near the cutoff.

2.6 FIT instrument

The flow-induced trading instrument uses mutual fund flows and lagged fund holdings. In simplified form,

$$FIT_{n,t} = \sum_i f_{i,t} S_{i,n,t-2}, \quad (7)$$

where $f_{i,t}$ is fund i 's flow and $S_{i,n,t-2}$ is its lagged ownership share of stock n .

Interpretation: The instrument predicts price pressure from mutual-fund flows into stocks the funds already held. Lagged holdings help isolate cross-sectional price pressure from contemporaneous stock-level cash-flow news.

2.7 Stock-level controls and samples

- Russell specifications focus on stocks around the Russell 1000/2000 reconstitution cutoff.
- Baseline BMI controls include log market capitalization, rolling average bid-ask spread, Russell 2000 indicator, rank-date market-cap rank, and banding controls.
- FIT specifications use stock fixed effects and quarter fixed effects.
- Robustness checks add observed and latent stock characteristics, minimum fund-holding screens, CRSP flags, passive-fund samples, and alternative fund subsets.

Interpretation: The controls are designed to ensure that the instruments move prices through nonfundamental demand pressure rather than through cash-flow news.

3 Models and empirical specifications

3.1 BMI two-stage least squares

The baseline BMI IV specification is

$$\Delta p_{a,n,t} = a_1 \Delta BMI_{n,t} + \beta'_1 X_{n,t} + FE_t + e_{1,a,n,t}, \quad (8)$$

$$\Delta E_{a,n,t+h|t+1} = a \widehat{\Delta p}_{a,n,t} + \beta'_2 X_{n,t} + FE_t + e_{2,a,n,t}. \quad (9)$$

- The first stage estimates the effect of benchmarking-intensity changes on price changes.
- The second stage estimates the effect of instrumented price changes on EPS expectation revisions.
- Standard errors are clustered by quarter.

Interpretation: The coefficient a is the causal effect of nonfundamental price changes on analyst cash-flow expectations.

3.2 FIT two-stage least squares

The FIT IV specification is

$$\Delta p_{a,n,t} = a_0 + a_1 FIT_{n,t} + FE_t + e_{1,a,n,t}, \quad (10)$$

$$\Delta y_{a,n,t} = b_0 + a \widehat{\Delta p}_{a,n,t} + FE_t + e_{2,a,n,t}, \quad (11)$$

where $\Delta y_{a,n,t}$ is either the LTG expectation change or an annual EPS expectation revision.

Interpretation: The FIT instrument provides an independent source of price variation and applies to much larger samples than the Russell cutoff design.

3.3 Term structure of annual EPS expectations

The paper estimates the FIT IV specification separately for annual EPS horizons $h = 1, 2, 3, 4$:

$$\Delta E_{a,n,t+h|t+1} = b_h + a_h \widehat{\Delta p}_{a,n,t} + FE_t + e_{2,a,n,t,h}. \quad (12)$$

Interpretation: If the price impact is merely a short-lived mistake, the effect should shrink at longer horizons. The paper finds that it does not.

3.4 Lagged FIT and reversal tests

To test whether analysts quickly reverse price-induced expectations, the paper estimates

$$\Delta y_{a,n,t} = \sum_{\ell=0}^L \beta_{\ell} FIT_{n,t-\ell} + FE_t + FE_n + e_{a,n,t}. \quad (13)$$

Interpretation: If analysts later learn that earlier price pressure was nonfundamental, lagged coefficients should reverse. The paper instead finds persistence.

3.5 Covariance decomposition

A key decomposition separates the covariance between analyst expectations and prices into a part explained by the causal impact of prices on analyst expectations and a residual common-shock part:

$$\frac{\sum_i Cov^{CX}(\alpha_h^{ATE} \Delta p_{a,n,t}^i, \Delta p_{a,n,t})}{Cov^{CX}(\Delta y_{a,n,t}^h, \Delta p_{a,n,t})}. \quad (14)$$

Interpretation: This asks how much of the observed price-expectation covariance is caused by prices changing analyst beliefs, rather than both being moved by cash-flow news.

3.6 Rational-investor model

The model contains:

- rational investors with private information;
- discount-rate variation that drives prices, excess volatility, and return predictability;
- analysts who learn from prices as signals of future cash flows;
- analyst expectations that become biased because prices contain discount-rate variation.

Interpretation: The model reconciles biased analyst beliefs with rational investors by separating analyst beliefs from investor beliefs.

4 Identification and empirical design

4.1 What identifies the price effect

- BMI and FIT generate price movements from institutional demand and fund-flow pressure.
- These price movements are designed to be unrelated to contemporaneous cash-flow news.
- The outcome is an analyst expectation revision over the same period in which the price change occurs.
- The IV estimates therefore identify whether price changes themselves move analyst cash-flow expectations.

Interpretation: The central identification assumption is that the instruments affect analyst expectations only through prices, not through future fundamentals or direct analyst information.

4.2 Threats addressed in the BMI design

- The paper uses May, not June, market capitalizations to compute Russell cutoffs, avoiding selection bias.
- It approximates proprietary Russell market caps to predict index assignment accurately.

- It checks that BMI increases do not predict future profitability or real effects.
- It varies bandwidths, winsorization rules, and controls.

Interpretation: These checks support the idea that BMI changes create price pressure rather than cash-flow news.

4.3 Threats addressed in the FIT design

- The instrument uses lagged mutual-fund holdings.
- It controls for observed stock characteristics and latent characteristics correlated with fund ownership.
- It reconstructs the instrument from diversified funds, passive funds, and alternative fund subsets.
- It tests whether the instrument forecasts profitability and finds limited evidence of fundamental effects.

Interpretation: The FIT design is more broad-based than the Russell design, but requires more controls because fund ownership patterns may correlate with firm characteristics.

4.4 Why the result matters for subjective beliefs data

- Analyst expectations correlate strongly with prices.
- Analyst expectations forecast future returns and forecast errors.
- These facts were previously viewed as evidence that investor beliefs are biased and distort prices.
- The paper shows that the same facts can arise if analysts learn from rational prices that contain discount-rate variation.

Interpretation: Subjective beliefs data do not necessarily reveal investor beliefs. Analysts and investors may have different information and different inference problems.

4.5 Limits of the design

- Analyst expectations are not direct measures of investor beliefs.
- The instruments isolate price pressure, but no instrument can fully rule out all indirect channels.
- The model is an example showing plausibility, not a proof that investors are always rational.
- The results leave open what economic forces generate the discount-rate variation in prices.

Interpretation: The paper does not reject biased investors. It shows that analyst belief moments alone are insufficient to conclude that investor beliefs are biased.

5 Tables and figures

Visuals are directly extracted from the paper and grouped compactly.

5.1 Figures 1–2: timing of analyst revisions and Russell reconstitution

Read:

- Figure 1 shows how analyst report dates define the price-change window and expectation-revision window.
- Figure 2 shows the Russell reconstitution design: analyst expectation changes are measured around the May rank date and June index change.

Economic message: Identification requires matching price changes to analyst belief revisions while avoiding cash-flow news in the same window.

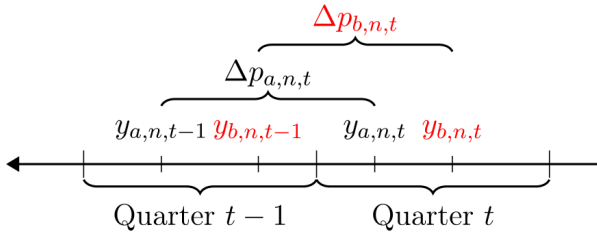


Figure 1: staggered analyst-report timing

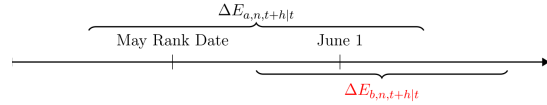


Figure 2: Russell reconstitution timeline

5.2 Tables 1–2: Russell BMI instrument and baseline IV result

Read:

- Table 1 summarizes the BMI sample: 164,512 analyst-stock-horizon-year observations in the 150-stock bandwidth.
- Table 2 shows that BMI changes strongly predict price changes; the first-stage coefficient is 0.573 and the partial F -statistic is 11.68.
- The reduced-form coefficient is 0.238.
- The 2SLS coefficient is 0.416, meaning a 1% instrumented price increase raises annual EPS expectation revisions by about 41 basis points.

Economic message: The Russell design provides clean evidence that price pressure unrelated to cash-flow news raises analyst cash-flow expectations.

Table 1
Summary Statistics for Russell Reconstitution Instrument.

	$\Delta BMI_{i,j}$	$\Delta E_{i,j,t+h t}$	$\Delta p_{i,j,t}$	Num Stocks/Year	Percent Covered
Num. Obs.	164,512.00	164,512.00	164,512.00	20.00	20.00
Mean	0.00	-0.02	0.02	432.45	0.89
Std. Dev.	0.04	0.20	0.24	148.33	0.04
Min	-0.41	-0.63	-0.60	240.00	0.80
25%	-0.01	-0.09	-0.12	260.00	0.87
50%	0.00	0.00	0.00	541.50	0.91
75%	0.01	0.07	0.13	554.00	0.92
Max	0.29	0.51	0.90	562.00	0.94

Summary statistics for observations in the 150-stock window around Russell reconstitution market cap thresholds in each year for May-to-June changes in benchmarking intensity ($\Delta BMI_{i,j}$), quarter-over-quarter revisions in analyst EPS expectations for forecast horizons of one to four years ($\Delta E_{i,j,t+h|t}$), inter-announcement percentage price changes between expectation releases in consecutive quarters ($\Delta p_{i,j,t}$), the number of stocks in the window in each year, and the percentage of all stocks in each window in each year that I observe analyst expectations for. Expectations revisions and price changes are expressed in absolute terms (i.e. 0.01 is 1%). The time period is 1999-05-2018-09.

Table 1: summary statistics for Russell instrument

Table 2
Impact of Prices on Annual EPS Expectations Using ΔBMI as Instrument.

	(1) OLS	(2) First Stage	(3) Reduced Form	(4) 2SLS
$\Delta p_{i,j,t}$	0.265*** (0.0111)			0.416** (0.0639, 0.768)
$\Delta BMI_{i,j}$		0.573*** (0.168)	0.238*** (0.0736)	
Year FE	Y	Y	Y	Y
Year-Clustered SE	Y	Y	Y	Y
N	164512	164512	164512	164512
F	568.6	11.68	10.48	13.95
R-Squared	0.0825	0.00747	0.00152	

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

This table displays results for the following two-stage least squares regression in the 150-stock window:

$$\Delta p_{i,j,t} = \alpha_0 + \alpha_1 \Delta BMI_{i,j} + \beta' X_{i,j} + \epsilon_{i,j,t}$$

The first stage regresses percent price changes between analyst reports ($\Delta p_{i,j,t}$) on the June change in BMI ($\Delta BMI_{i,j}$) (column 2 reports the first-stage partial F -statistic). The reduced form regresses quarterly revisions to annual EPS expectations with horizons of one to four years ($\Delta E_{i,j,t+h|t}$) on $\Delta BMI_{i,j}$. The second stage regresses $\Delta E_{i,j,t+h|t}$ on instrumented price changes ($\Delta p_{i,j,t}$). $X_{i,j}$ includes the log market cap as of the May rank date, the one-year monthly rolling average bid-ask percentage spread, and the banding controls (an indicator for having rank-date market cap in the band including stocks from the Russell 1000 and 2000, an indicator for being in the Russell 2000 in May before reconstitution, and the interaction of these indicators). Column 4 reports the 95% confidence interval for the second-stage coefficient using the rF procedure of Lee et al. (2022). All units are in percentage points (i.e. 1.0 is 1%). The time period is 1999-05-2018-09.

Table 2: BMI IV results

5.3 Figure 3: BMI robustness checks

Read:

- Figure 3 reports first-stage, reduced-form, and second-stage estimates across alternative specifications.
- Robustness checks include no price winsorization, alternative winsorization of earnings expectations, alternative bandwidths, value controls, and a post-rank-date sample.
- The second-stage estimates remain broadly similar to the baseline.

Economic message: The BMI result is not driven by a single bandwidth, winsorization choice, or control set.

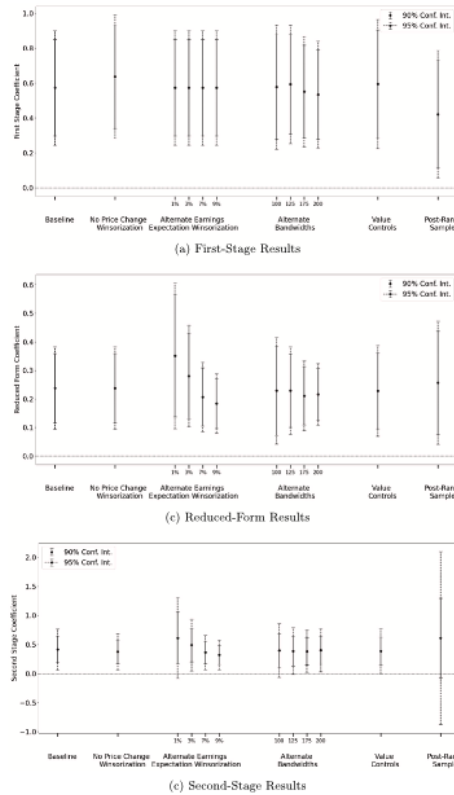


Fig. 3. Robustness checks for using $ARNFI$ as instrument. This figure displays results for alternate specifications of two-stage least squares regression (5). In all specifications X_{it} includes the log market cap as of the May rank date, the one-year monthly rolling average bid-ask percentage spread, and the banding controls (an indicator for having rank-date market cap in the band including stocks from the Russell 1000 and 2000, an indicator for being in the Russell 2000 in May before reconstitution, and the intersection of these indicators). In the "Value Controls" specification, X_{it} also includes market-to-book ratio and annual sales growth. The solid error bars display 90% confidence intervals, while the dashed error bars display 95% confidence intervals. Panel (c) reports the 90% and 95% confidence intervals for the second-stage coefficients using the IF procedure of [Lee et al. \(2022\)](#). Standard errors are clustered by quarter. All units are in percentage points (i.e. 1.0 is 1%). The time period is 1999-05-2018-09.

5.4 Tables 3–4: FIT instrument and baseline IV result

Read:

- Table 3 summarizes the FIT sample for LTG expectations and annual EPS expectations.
- Table 4 shows a strong first stage: flow-induced trading predicts price changes.
- In Panel A, a 1% instrumented price increase raises LTG expectations by 5.46 basis points.
- In Panel B, a 1% instrumented price increase raises annual EPS expectation revisions by 20.6 basis points.

Economic message: FIT confirms the main price-to-belief channel in a much larger sample and with a different source of price pressure.

Table 3
Summary Statistics for FIT Instrument.

	$\Delta LTG_{i,t}$	$\Delta p_{i,t}$	$FIT_{i,t}$	Num Stocks/(Fund, Quarter)
Num. Obs.	121 553.00	121 553.00	121 553.00	131 333.00
Mean	-0.01	0.03	-0.0001	165.30
Std. Dev.	0.04	0.22	0.0039	309.02
Min	-0.12	-0.94	-0.3941	1.00
25%	-0.02	-0.06	-0.0010	44.00
50%	-0.00	0.02	-0.0000	73.00
75%	0.01	0.13	0.0007	132.00
Max	0.10	5.80	0.1845	3712.00

(a) LTG Expectations	$\Delta E_{i,t+4 t}$	$\Delta p_{i,t}$	$FIT_{i,t}$	Num Stocks/(Fund, Quarter)
Num. Obs.	3396550.00	3396550.00	3396550.00	133 902.00
Mean	-0.01	0.03	-0.0001	162.61
Std. Dev.	0.19	0.24	0.0042	306.68
Min	-0.59	-0.99	-0.4536	1.00
25%	-0.08	-0.09	-0.0011	43.00
50%	0.00	0.02	-0.0000	72.00
75%	0.07	0.14	0.0008	130.00
Max	0.44	14.05	0.2134	3712.00

(b) Annual EPS Expectations — All Horizons	$\Delta E_{i,t+4 t}$	$\Delta p_{i,t}$	$FIT_{i,t}$	Num Stocks/(Fund, Quarter)
Num. Obs.	3396550.00	3396550.00	3396550.00	133 902.00
Mean	-0.01	0.03	-0.0001	162.61
Std. Dev.	0.19	0.24	0.0042	306.68
Min	-0.59	-0.99	-0.4536	1.00
25%	-0.08	-0.09	-0.0011	43.00
50%	0.00	0.02	-0.0000	72.00
75%	0.07	0.14	0.0008	130.00
Max	0.44	14.05	0.2134	3712.00

Summary statistics for quarter-over-quarter changes in LTG expectations $\Delta LTG_{i,t}$ (Panel (a)) and revisions in annual EPS expectations for

Table 3: summary statistics for FIT instrument

5.5 Figures 4–5: term structure and persistence

Read:

- Figure 4 shows that the effect of prices on annual EPS expectations does not shrink with the forecast horizon from one to four years.
- Figure 5 shows lagged FIT coefficients: the impact does not immediately reverse in the next year.
- The cumulative effect persists for LTG and annual EPS expectation revisions.

Economic message: Analysts do not simply correct a short-run mistake once price pressure becomes public. Price-induced expectation changes are persistent.

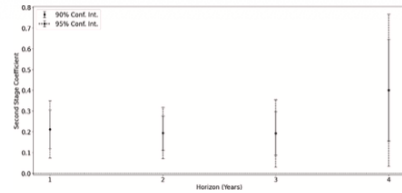


Fig. 4. Impact of prices on term structure of EPS expectations. This figure displays the following two-stage least squares regressions results:
 $\Delta p_{i,t} = \alpha_0 + \alpha_1 FIT_{i,t} + FE_i + \epsilon_{i,t}$
 $\Delta E_{i,t+4|t} = \beta_0 + \beta_1 \Delta p_{i,t} + FE_i + \epsilon_{i,t+4|t}$
The first stage regresses quarterly percent price changes ($\Delta p_{i,t}$) on the flow-induced trading instrument ($FIT_{i,t}$). The second stage regresses quarterly revisions to annual EPS expectations ($\Delta E_{i,t+4|t}$) on the instrumented price changes ($\Delta p_{i,t}$). I run this regression separately for each horizon h from 1 to 4 years. FE_i are quarter fixed effects. All units are in percentage points (i.e. 1.0 is 1%). The solid error bars represent 90% confidence intervals, while the dashed error bars display 95% confidence intervals, both of which are computed using the IF procedure of Lee et al. (2022). Standard errors are clustered by quarter. The time period is 1982-04-2020-12.

Figure 4: term structure of EPS expectation effects

Table 4
Impact of Prices on Analyst Expectations Using FIT as Instrument.

	(1)	(2)	(3)	(4)
Panel (a): LTG Expectation Changes	OLS	First Stage	Reduced Form	2SLS
$\Delta p_{i,t}$	0.0438*** (0.00209)			0.0546** (0.0169, 0.0923)
$FIT_{i,t}$		3.355*** (0.920)	0.183*** (0.0602)	
N	121 553	121 553	121 553	121 553
F	439.5	13.31	9.242	18.84
R-Squared	0.0399	0.00373	0.000231	

Panel (b): Revisions of Annual EPS Expectations				0.206** (0.0820, 0.330)
$\Delta p_{i,t}$	0.279*** (0.0118)			
$FIT_{i,t}$		3.463*** (0.8603)	0.714*** (0.203)	
N	3 396 550	3 396 550	3 396 550	3 396 550
F	862.2	16.09	12.32	21.51
R-Squared	0.108	0.00387	0.000229	
Quarter FE	Y	Y	Y	Y
Quarter-Clustered SE	Y	Y	Y	Y

Standard errors in parentheses
 $^* p < 0.10$, $^{**} p < 0.05$, $^{***} p < 0.01$
This table displays results for the following two-stage least squares regression:

$$\Delta p_{i,t} = \alpha_0 + \alpha_1 FIT_{i,t} + FE_i + \epsilon_{i,t}$$

$$\Delta E_{i,t+4|t} = \beta_0 + \beta_1 \Delta p_{i,t} + FE_i + \epsilon_{i,t+4|t}$$

The first stage regresses percent price changes between analyst reports ($\Delta p_{i,t}$) on the flow-induced trading instrument ($FIT_{i,t}$) (column 2 reports the first-stage partial F -statistic). The reduced form regresses changes in LTG expectations ($\Delta LTG_{i,t}$) in Panel (a) and quarterly revisions to annual EPS expectations with horizons of one to four years ($\Delta E_{i,t+4|t}$) in Panel (b) on $FIT_{i,t}$. The second stage regresses $\Delta LTG_{i,t}$ in Panel (a) and $\Delta E_{i,t+4|t}$ in Panel (b) on the instrumented price changes ($\Delta p_{i,t}$). FE_i are quarter fixed effects. All units are in percentage points (i.e. 1.0 is 1%). Column 4 reports the 95% confidence interval for the second-stage coefficients using the IF procedure of Lee et al. (2022). The time period is 1983-01-2020-12 for the LTG expectations and 1982-04-2020-12 for the annual EPS expectation revisions.

price-induced tradline raises annual EPS expectation revisions. Do analysts revise their expectations ex-post once they hear

Table 4: FIT IV results

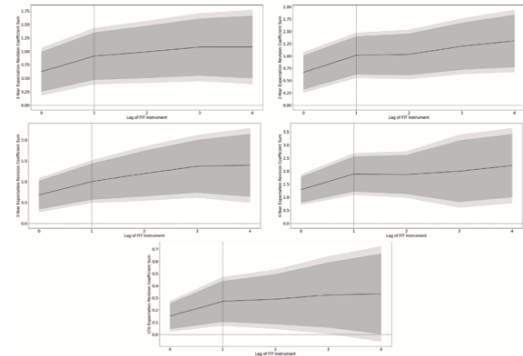


Fig. 5. Lack of immediate reversal in impact of FIT on analyst cash flow expectations. This figure displays the coefficient sums $\sum_{l=0}^L \beta_l$, $L = 0, \dots, 4$, from the following regression:

$$\Delta E_{i,t+L+1|t} = \sum_{l=0}^L \beta_l FIT_{i,t-l} + FE_i + \epsilon_{i,t+L+1|t}$$

$\Delta E_{i,t+L+1|t}$ is either the quarter-over-quarter change in LTG expectations for analyst institution i for stock i in quarter t ($\Delta LTG_{i,t}$), or the quarterly revision to annual EPS expectations with horizons of one to four years ($\Delta E_{i,t+L+1|t}$). FE_i and FE_t are quarter and stock fixed effects. Dark and light shaded areas represent 90% and 95% confidence intervals, respectively. Standard errors are clustered by quarter and stock. All units are in percentage points (i.e. 1.0 is 1%). The time period is 1983-01-2020-12 for the LTG expectations and 1982-04-2020-12 for the annual EPS expectation revisions.

Figure 5: persistence of lagged FIT effects

5.6 Figure 6 and Table 5: FIT robustness and Rotemberg weights

Read:

- Figure 6 reports robustness checks for the FIT instrument.
- Results are similar after adding observed characteristics, latent characteristics, minimum holding thresholds, CRSP flags, and index-fund-only screens.
- Table 5 shows the most important fund styles by Rotemberg weight; domestic equity styles dominate.

Economic message: The FIT estimates are not mechanically generated by one particular group of funds or by obvious stock characteristics.

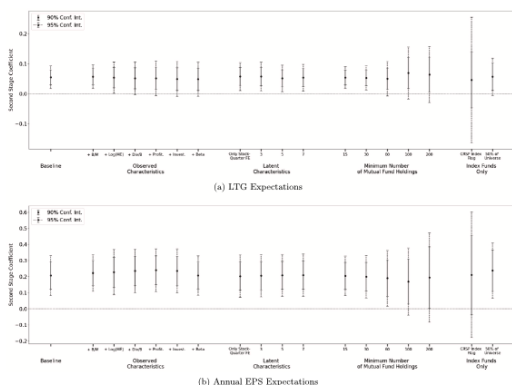


Fig. 6. Robustness checks for using FIT instrument.

This figure displays the following two-stage least squares regression results:

$$AP_{i,t+M} = \alpha_0 + \alpha_1 FIT_{i,t} + \beta_1 X_{i,t} + FE_{i,t} + \epsilon_{i,t+M}$$

The first stage regresses percent price changes between analyst reports ($AP_{i,t}$) on the flow-induced trading instrument ($FIT_{i,t}$). The second stage regresses changes in LTG expectations ($\Delta LTG_{i,t+M}$) in Panel (a) and quarterly revisions to annual EPS expectations with horizons of one to four years ($\Delta EPS_{i,t+M}$) in Panel (b) on the instrumented price changes ($\Delta AP_{i,t}$). $X_{i,t}$ includes quarter indicators interacted with either observed (in the "Observed Characteristics" specification) or latent (in the "Latent Characteristics" specification) stock characteristics. The observed stock characteristics include: book-to-market ratio, log market equity, dividend-to-book equity ratio, profitability, investment, and market beta. For the "Observed Characteristics" specification, each subsequent column adds an additional control variable (e.g. the right-most column represents the results of the regression with all six control variables). The time period for the "Observed Characteristics" and "Latent Characteristics" specifications is 1983-01-2020-12 for the LTG expectations and 1982-04-2020-12 for the annual EPS expectation revisions. For the "Minimum Number of Mutual Fund Holdings" specification, each column reconstructs $FIT_{i,t}$ only from mutual funds that have at least M holdings, where M is labeled on the x-axis. The time period for the "Minimum Number of Mutual Fund Holdings" specifications is 1983-01-2020-12 for the LTG expectations and 1982-04-2020-12 for the annual EPS expectation revisions. The "Index Funds Only" specifications reconstruct $FIT_{i,t}$ from only mutual funds identified as index funds by one of the two criteria listed on the x-axis. The time period for the "Index Funds Only" specifications is 1984-09-2020-12. $FE_{i,t}$ are quarter fixed effects. The solid error bars represent 95% confidence intervals, while the dotted error bars display 95% confidence intervals, both of which are computed using the iv procedure of *Lee et al. (2022)*. Standard errors are clustered by quarter. All units are in percentage points (i.e. 1.0 is 1%).

Table 5

Rotemberg Weights as Percentage of Total for Most Important Fund Styles.

Style	LTG	Annual EPS Expectation
Domestic Equity Growth	82.74	38.08
Domestic Equity Small Cap	27.23	35.97
Domestic Equity Income	3.98	10.20
Domestic Equity Growth & Income	2.37	4.08

Total Rotemberg weights (expressed as percentages of the total weight) for top four most important fund styles for both the LTG expectations ($\Delta LTG_{i,t+M}$) and annual EPS expectations ($\Delta EPS_{i,t+M}$) samples.

Figure 6: FIT robustness checks

Table 5: Rotemberg weights by fund style

5.7 Figure 7: covariance decomposition

Read:

- Figure 7 decomposes the covariance between analyst cash-flow expectations and prices.
- The red bars show the portion explained by the causal impact of prices on analyst expectations.
- The gray bars show the portion explained by common information or sentiment shocks to analyst and investor expectations.
- The price-to-expectations channel explains a large share of the covariance for both LTG and one-to-four-year annual EPS expectations.

Economic message: The beliefs-data covariance with prices is not purely evidence that biased analyst-like beliefs move prices. A large part may reflect analysts learning from prices.

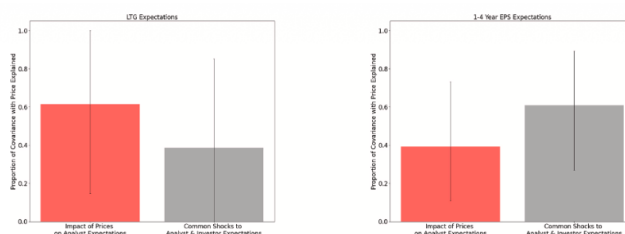


Fig. 7. Decomposition of covariance between analyst cash flow expectations and prices.

This figure displays the proportion of the covariance of analyst cash flow expectation changes with contemporaneous price changes explained by the impact of prices on analyst cash flow expectations (red) and by common information or sentiment shocks to analyst and investor expectations (gray). These proportions sum to one. The left and right panels display this decomposition for changes in LTG expectations and revisions to one to four year EPS expectations (which capture forecast error changes). Error bars represent quarterly block-bootstrapped 95% confidence intervals.

5.8 Table 6: model estimation

Read:

- Table 6 reports externally calibrated parameters, estimated parameters, and target/model moments.

- The model matches cross-sectional annual EPS growth variance, EPS growth autocorrelation, market-to-book autocorrelation, return predictability, analyst forecast-error predictability, and the impact of prices on analyst expectations.
- Investors are rational, but analysts form biased cash-flow expectations because they learn from prices that include discount-rate variation.

Economic message: The model shows that rational-investor asset pricing can reproduce key belief moments once analyst learning from prices is incorporated.

Panel (a): Externally Calibrated Parameters			
Parameter	Value		
r_f	0.043		
σ	0.23		
μ_d	0.063		
Panel (b): Estimated Parameters			
Parameter	Value		
τ	1.73		
σ_{pub}	0.02		
σ_{priv}	0.001		
σ_f	0.03		
σ_e	0.30		
ρ_{pub}	0.48		
ρ_{priv}	0.89		
ϕ	0.91		
α	1.88		
Panel (c): Target and Model Moments			
	Moment	Target	Model
$V[G_{it}]$	Cross-Sectional Annual EPS Growth Variance	0.36	0.36
$\frac{Cov(G_{it}, G_{it-1})}{V[G_{it}]}$	1st Order Annual EPS Growth Autocorrelation	0.70	0.75
$\frac{Cov(G_{it}, G_{it-2})}{V[G_{it}]}$	2nd Order Annual EPS Growth Autocorrelation	0.37	0.50
$\frac{Cov(log P_{it}, log P_{it-1})}{V[log P_{it}]}$	1st Order Valuation Ratio Autocorrelation	0.64	0.63
$log E[P_{it} - r_f]$	Log Equity Premium	0.06	0.09
$\frac{Cov(log P_{it}, log P_{it-1})}{V[log P_{it}]}$	Level of Cross-Sectional Return Predictability	0.20	0.19
$\frac{Cov(log E_{it}, log E_{it-1})}{V[log E_{it}]}$	Analyst Forecast Error Predictability	-0.01	-0.03
$\frac{Cov(log P_{it}, log P_{it-1})}{V[log P_{it}]}$	Return Predictability by Analyst Expectations	-0.15	-0.18
$\frac{Cov(log P_{it}, log P_{it-1})}{V[log P_{it}]}$	Covariance of Subjective Expected Returns with Prices	0	0
$\frac{Cov(log P_{it}, log P_{it-1})}{V[log P_{it}]}$	Impact of Prices on Analyst Cash Flow Expectations	0.12	0.11

This table reports the externally calibrated parameters in Panel (a), the estimated parameters in Panel (b), and the targeted and model-implied moments in Panel (c). G_{it} is four-year dividend growth ($G_{it} = \sum_{j=1}^4 \Delta d_{i,t+j}$). κ_1 is the log-linearization constant from Campbell and Shiller (1988) ($\kappa_1 = 1/(1 + \exp[\ln(D_{it}/P_{it})])$). Appendix H.2 provides details.

6 Short summary

- The paper asks whether subjective beliefs data necessarily contradict rational-expectations discount-rate models.
- Analyst cash-flow expectations are strongly correlated with prices and forecast returns, which prior work interprets as evidence of biased investor expectations.
- Chaudhry shows that prices themselves affect analyst cash-flow expectations.
- The first IV uses Russell benchmarking-intensity changes around index reconstitutions.
- The second IV uses mutual fund flow-induced trading.
- Both instruments show that exogenous price increases raise analyst cash-flow expectations.
- The effect persists across forecast horizons and does not quickly reverse.
- The price-to-expectation channel explains a large part of the covariance between prices and analyst expectations.
- A rational-investor model with private information and analysts learning from prices can match key subjective-belief moments.
- Analyst expectations are therefore useful data, but they should not be mechanically interpreted as investor expectations.

7 Conclusion

7.1 Core conclusion

Chaudhry shows that prices have a causal effect on analyst cash-flow expectations. This finding changes the interpretation of subjective beliefs evidence: biased analyst expectations that correlate with prices and forecast returns do not necessarily imply that investors have biased expectations that distort prices.

7.2 Economic intuition

Analysts use prices as signals. If prices contain both cash-flow information and discount-rate variation, analysts who learn from prices can mistakenly interpret discount-rate-driven price movements as cash-flow news. Hence analyst cash-flow expectations can become biased even when investors are rational and prices are driven by discount-rate variation.

7.3 Takeaway

Subjective beliefs data should be interpreted through an information structure. The paper's main lesson is not that investors are certainly rational, but that analyst beliefs alone cannot prove that investor beliefs are biased. Prices can shape beliefs, and this feedback is central for interpreting belief data in asset pricing.